



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to BOOMALAIYAPPA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

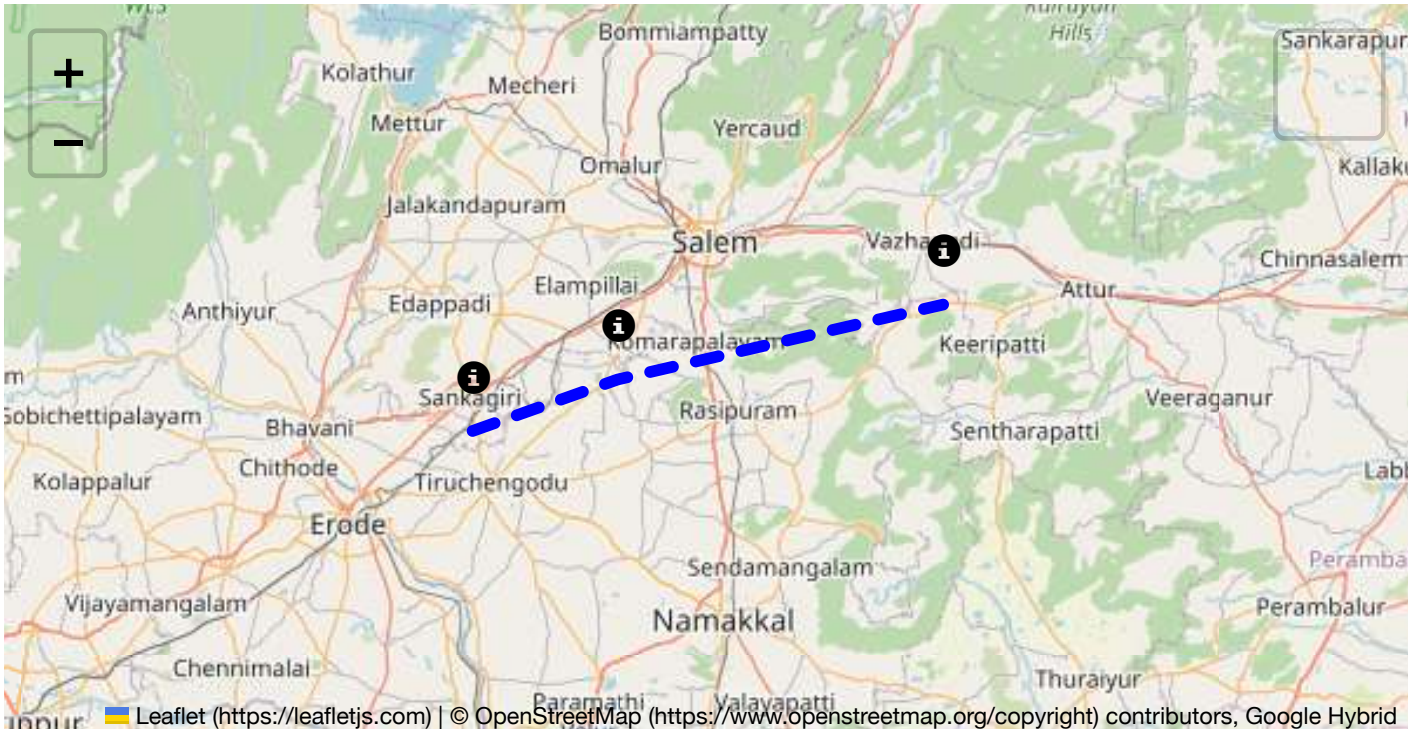
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 77.83 km
Estimated Duration: 2.0 hours
Adjusted Duration (Heavy Vehicle): 2.5 hours

Start: (11.4381, 77.8734)
End: (11.583028, 78.424991)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route travels through rural and semi-urban areas in Tamil Nadu, covering approximately 77.83 kilometers. It generally follows a north-northwest path, passing through the town of Periya Kollappatty. The main roadways used include regional highways and local roads that connect smaller communities to larger state highways.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate with hot summers, mild winters, and a monsoon season from June to September. During monsoons, heavy rainfall can lead to flooding and waterlogging on roads, increasing the risk of hydroplaning and accidents. Fog can be present during early winter mornings, reducing visibility.

3. Analysis of Traffic Patterns

Traffic congestion is typically experienced in the mornings (8:00-10:00 AM) and evenings (5:00-8:00 PM), particularly around town areas like Sangagiri and Periya Kollappatty. Roads near market areas and schools can also have sudden slowdowns due to pedestrian and two-wheeler traffic.

4. Assessment of Road Quality and Infrastructure

Road quality varies throughout the route. Highways are generally well-paved but secondary roads and those passing through towns may have potholes and uneven surfaces. Some areas lack sufficient lighting, posing risks during night travel. Signage may not be adequate in certain rural segments.

5. Suggestions for Alternative Routes for Emergencies

In case of roadblocks or emergencies, NH 44 is a viable alternative since it runs parallel to many parts of the planned route and offers better facilities. It would require a detour but is more reliable during emergencies.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to regulations concerning loading, vehicle markings, and safety equipment. Permits are needed, and transport is restricted during peak traffic times and in sensitive areas. All drivers must carry a hazardous materials transportation license.

7. Overview of Historical Incidents

There have been occasional accidents involving heavy vehicles, typically due to driver error or poor road conditions, but no major hazardous material incidents reported. Local roads with sharp turns and insufficient signage have been highlighted as risk factors in past reports.

8. Environmental Considerations and Sensitive Areas

The route passes near agricultural lands, which require careful hazard material transport to prevent contamination. Additionally, local water bodies should be protected against spills. There are no major wildlife reserves along the primary route.

9. Analysis of Communication Coverage

Mobile coverage is generally good but can be spotty in rural stretches and hilly areas. It is advisable to have multiple networks available or use satellite communication devices if consistent communication is necessary.

10. Estimated Emergency Response Times

Emergency response times vary. Urban areas can expect 20-30 minute response times from emergency services. Rural segments may face delays up to 60 minutes due to the distance and road quality. It is crucial to plan for emergencies through communication and coordinated action plans.

12. Overall Summary of Risk Assessment

The primary risk factors include adverse weather during the monsoon, road quality issues, and potential delays in emergency response. Adequate planning and risk mitigation measures such as alternative routes, emergency preparedness, and communication solutions should be implemented. With proper precautions, the route is manageable, but continuous monitoring for changes in local conditions and regulations is recommended.

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.46154, 78.18539	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44833, 77.87437	30 KM/Hr
4	Turn	Medium	11.44868, 77.87458	30 KM/Hr
5	Turn	Medium	11.45488, 77.88853	30 KM/Hr
6	Turn	Medium	11.45452, 77.88964	30 KM/Hr
7	Blind Spot	Blind Spot	11.43684, 77.91706	10 KM/Hr
8	Turn	High	11.43913, 77.91702	15 KM/Hr
9	Turn	Medium	11.43979, 77.93009	30 KM/Hr
10	Turn	Medium	11.44030, 77.93066	30 KM/Hr
11	Turn	Medium	11.44088, 77.93223	30 KM/Hr
12	Turn	High	11.44243, 77.93244	15 KM/Hr
13	Turn	High	11.44285, 77.93440	15 KM/Hr
14	Turn	Medium	11.44100, 77.93748	30 KM/Hr
15	Turn	Medium	11.44543, 77.95299	30 KM/Hr
16	Turn	Medium	11.44468, 77.95810	30 KM/Hr
17	Turn	Medium	11.44395, 77.96298	30 KM/Hr
18	Turn	High	11.49058, 78.02690	15 KM/Hr
19	Turn	High	11.49138, 78.03235	15 KM/Hr
20	Turn	Medium	11.49190, 78.03281	30 KM/Hr
21	Turn	Medium	11.49409, 78.03886	30 KM/Hr
22	Turn	High	11.49641, 78.04988	15 KM/Hr
23	Turn	High	11.49672, 78.05001	15 KM/Hr
24	Blind Spot	Blind Spot	11.49723, 78.05308	10 KM/Hr
25	Turn	High	11.49546, 78.05344	15 KM/Hr
26	Turn	High	11.49545, 78.05906	15 KM/Hr
27	Turn	High	11.49467, 78.05909	15 KM/Hr
28	Turn	High	11.49446, 78.05997	15 KM/Hr
29	Blind Spot	Blind Spot	11.49303, 78.05992	10 KM/Hr
30	Turn	High	11.49375, 78.06145	15 KM/Hr
31	Turn	High	11.49563, 78.06197	15 KM/Hr
32	Turn	High	11.49666, 78.06789	15 KM/Hr
33	Turn	High	11.49728, 78.06789	15 KM/Hr
34	Turn	High	11.49740, 78.06877	15 KM/Hr
35	Turn	High	11.49913, 78.06940	15 KM/Hr
36	Turn	Medium	11.49950, 78.07090	30 KM/Hr
37	Blind Spot	Blind Spot	11.50121, 78.07189	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
38	Turn	High	11.50068, 78.07327	15 KM/Hr
39	Turn	High	11.50082, 78.07865	15 KM/Hr
40	Turn	Medium	11.50378, 78.08052	30 KM/Hr
41	Turn	High	11.50435, 78.08045	15 KM/Hr
42	Turn	Medium	11.51305, 78.08910	30 KM/Hr
43	Turn	Medium	11.51428, 78.08911	30 KM/Hr
44	Turn	High	11.51450, 78.08927	15 KM/Hr
45	Turn	Medium	11.51414, 78.09067	30 KM/Hr
46	Turn	Medium	11.51470, 78.09150	30 KM/Hr
47	Blind Spot	Blind Spot	11.51500, 78.09169	10 KM/Hr
48	Turn	Medium	11.51378, 78.09221	30 KM/Hr
49	Turn	Medium	11.49698, 78.14266	30 KM/Hr
50	Turn	Medium	11.48968, 78.14728	30 KM/Hr
51	Turn	High	11.48709, 78.15385	15 KM/Hr
52	Turn	High	11.47225, 78.15722	15 KM/Hr
53	Turn	Medium	11.46928, 78.17291	30 KM/Hr
54	Turn	Medium	11.46704, 78.17520	30 KM/Hr
55	Turn	High	11.46500, 78.18149	15 KM/Hr
56	Turn	High	11.46299, 78.18217	15 KM/Hr
57	Turn	Medium	11.45957, 78.18536	30 KM/Hr
58	Turn	High	11.45898, 78.18564	15 KM/Hr
59	Turn	Medium	11.45823, 78.18839	30 KM/Hr
60	Turn	Medium	11.44993, 78.22152	30 KM/Hr
61	Turn	Medium	11.44951, 78.22263	30 KM/Hr
62	Turn	High	11.44797, 78.22314	15 KM/Hr
63	Turn	Medium	11.44756, 78.22414	30 KM/Hr
64	Turn	Medium	11.45353, 78.23584	30 KM/Hr
65	Turn	Medium	11.49835, 78.32417	30 KM/Hr
66	Turn	Medium	11.50506, 78.33087	30 KM/Hr
67	Turn	Medium	11.50865, 78.33966	30 KM/Hr
68	Turn	Medium	11.51183, 78.34416	30 KM/Hr
69	Turn	Medium	11.51870, 78.34816	30 KM/Hr
70	Turn	Medium	11.54842, 78.35627	30 KM/Hr
71	Turn	Medium	11.56719, 78.37813	30 KM/Hr
72	Turn	Medium	11.56940, 78.39856	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital, Vennandur	11.51701, 78.088631	30 km/h	Medium
5	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
6	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
7	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
8	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
9	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
10	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium
11	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
12	hospital	R K Eye Care Center	11.4589079, 78.1913596	30 km/h	Medium
13	hospital	Manickkam Hospital	11.4568944, 78.1926061	30 km/h	Medium
15	hospital	Snathi Hospital	11.4564245, 78.1932485	30 km/h	Medium
16	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium
17	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium
4	school	SRV School	11.4688166, 78.1634582	30 km/h	Medium
14	school	Govt. Hr. Sec. School	11.4585324, 78.1933666	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46154, 78.18539



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr
Coordinates: 11.44833, 77.87437



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44868, 77.87458



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45488, 77.88853



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45452, 77.88964



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.43684, 77.91706



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43913, 77.91702



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43979, 77.93009



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44030, 77.93066



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44088, 77.93223



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44243, 77.93244



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44285, 77.93440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44100, 77.93748



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44543, 77.95299



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44468, 77.95810



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44395, 77.96298



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49058, 78.02690



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49138, 78.03235



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49190, 78.03281



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49409, 78.03886



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49641, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49672, 78.05001



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49723, 78.05308



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49546, 78.05344



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49545, 78.05906



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49467, 78.05909



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49446, 78.05997



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49303, 78.05992



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49375, 78.06145



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49563, 78.06197



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49666, 78.06789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49728, 78.06789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49740, 78.06877



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49913, 78.06940



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49950, 78.07090



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.50121, 78.07189



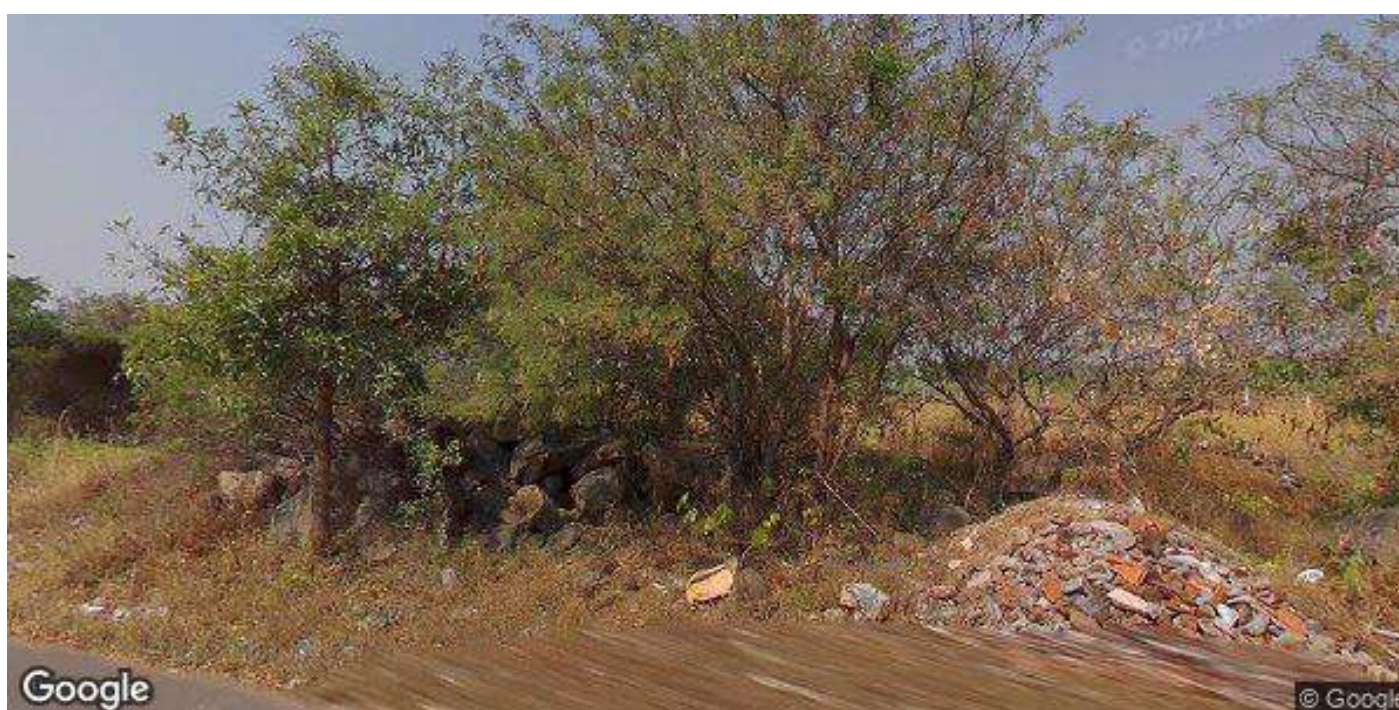
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50068, 78.07327



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50082, 78.07865



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50378, 78.08052



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50435, 78.08045



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51305, 78.08910



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51428, 78.08911



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51450, 78.08927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51414, 78.09067



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51470, 78.09150



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51500, 78.09169



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51378, 78.09221



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49698, 78.14266



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48968, 78.14728



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48709, 78.15385



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47225, 78.15722



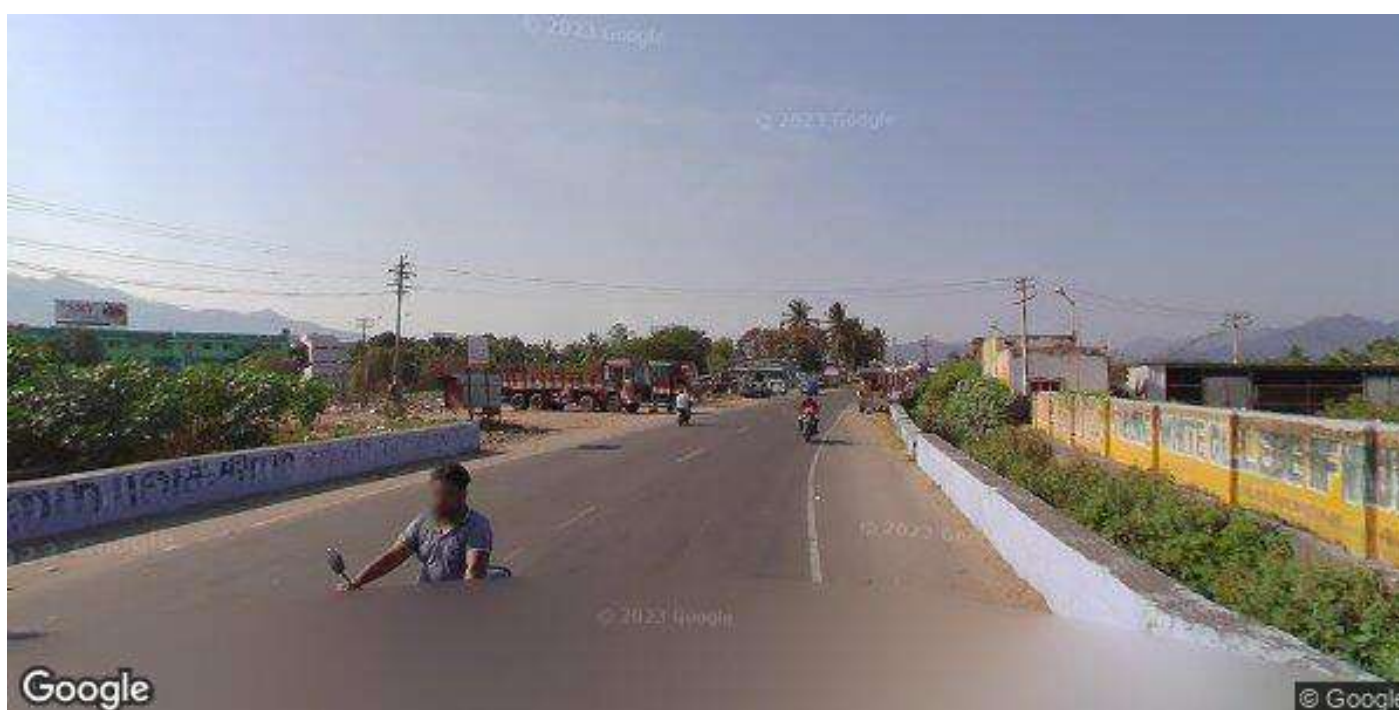
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46928, 78.17291



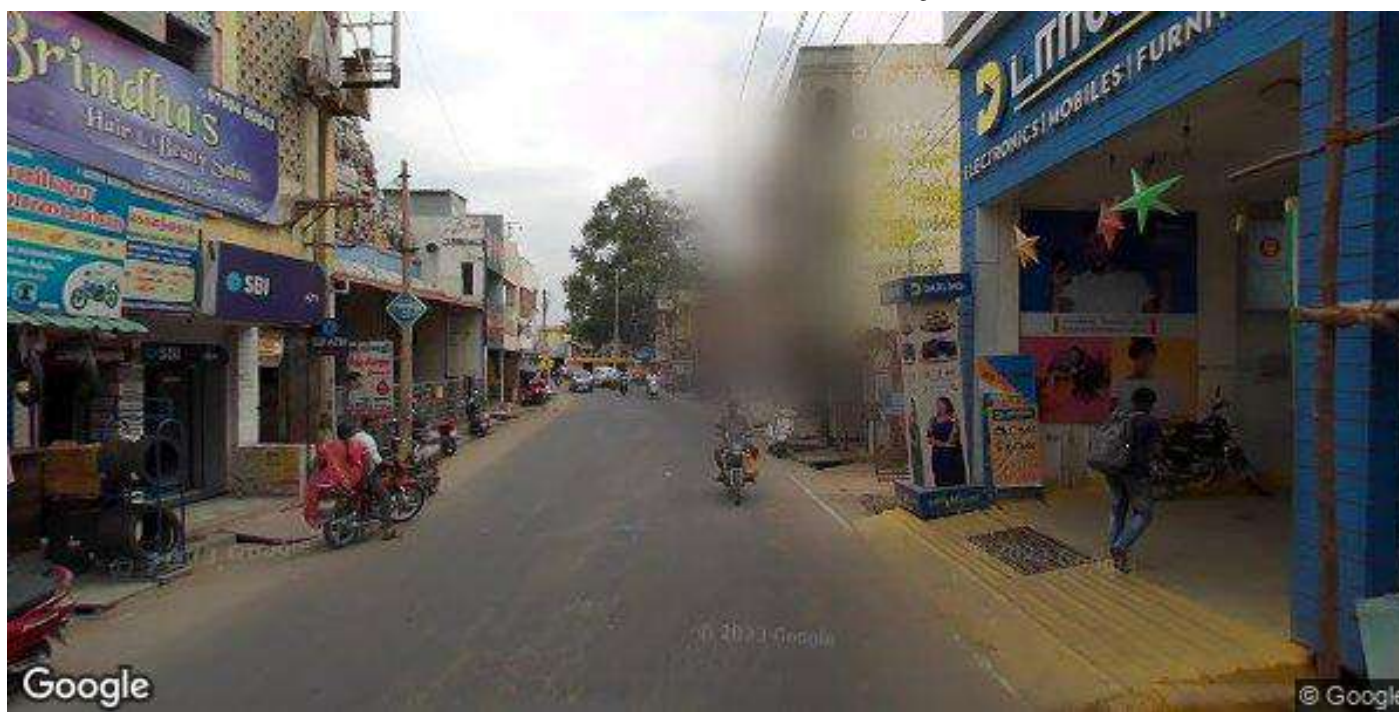
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46704, 78.17520



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46500, 78.18149



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46299, 78.18217



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45957, 78.18536



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45898, 78.18564



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45823, 78.18839



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44993, 78.22152



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44951, 78.22263



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44797, 78.22314



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44756, 78.22414



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 78.23584



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49835, 78.32417



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50506, 78.33087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50865, 78.33966



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51183, 78.34416



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51870, 78.34816



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54842, 78.35627



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56719, 78.37813



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56940, 78.39856
